

ARAB YOUTH SPACE HACKATHON 2026 · CHALLENGE 813 · TEAM T0049

QAYDH القيث

Heat-risk intelligence for fast-growing Gulf cities

Where the city grew · which districts burn hottest · **which materials** make them hot · **who is outside** in that heat. One map that tells planners where to act first, and with what.

Theme: Sustainable Urban Planning & Smart Cities, Urban Expansion, Land Use Change & Heat Risk · SDG 3 · 11 · 13

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Gulf summers are getting hotter, and people still have to be outside

- Gulf cities are among the fastest-growing on Earth. Surfaces pass **50 °C** in summer (median LST here: **52.2 °C** in established districts).
- Every new district locks in its roofs, roads and greenery for decades.
- Planners decide cool roofs, trees and shade **case by case**. There is no routine city-wide view linking **growth** → **heat** → **materials** → **people**.



Walking to Dhuhr & Asr
prayers at peak heat



Waiting at bus stops
often unshaded



Delivery riders
waiting outside restaurants



Construction workers
outdoors all day

User: municipal planning & heat-resilience offices · **Decision:** which neighbourhoods get cool roofs, shade trees, shaded bus stops, mosque-walkway shade or rider cooling points *first*

Five open-data layers, one decision map



Heat-Risk Priority Index = **heat** × **people exposure** × **lack of greenery** × **new-district factor**, so each hotspot comes with a named intervention: cool roofs, shade trees, shaded bus shelters, mosque-walkway shade, rider cooling points, or midday work-break enforcement.

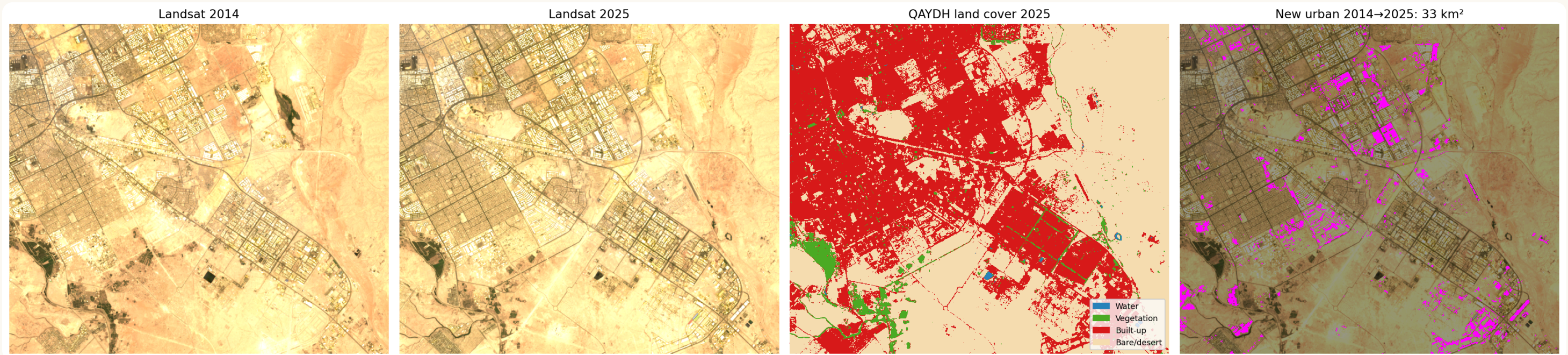
Portable by design: the Tanager footprint sets the area. Change one scene ID and the whole pipeline moves to another city. AOI here: east Riyadh (As Sali), 1,091,324 residents, 470 km².

Open data only, fully reproducible

Dataset	Use	Licence
Planet Tanager hyperspectral SR (open archive, 15 May 2025)	Materials, albedo, hyperspectral value test	CC-BY-4.0
Landsat 8/9 C2 L2 (SR + surface temperature)	Land cover 2014/17/21/23/25, summer LST	Public domain
ESA WorldCover 2021	Training labels (no manual annotation needed)	CC-BY-4.0
Impact Observatory LULC 2017 / 2023	Independent validation	CC-BY-4.0
WorldPop 2025 (100 m)	Residents per cell	CC-BY-4.0
OpenStreetMap	Mosques, bus stops, rider hubs, construction, schools, clinics	ODbL

Python · Planetary Computer STAC · scikit-learn · one Colab notebook · every scene ID in `data_provenance.json` · no imagery in GitHub. Next: **glQ** + Satellite 813 / MBZ-SAT in incubation.

East Riyadh grew 167 → 214 km² (+28%) since 2014



0.89

built-up F1, spatial-block CV
vs 0.58 for the official NDBI rule

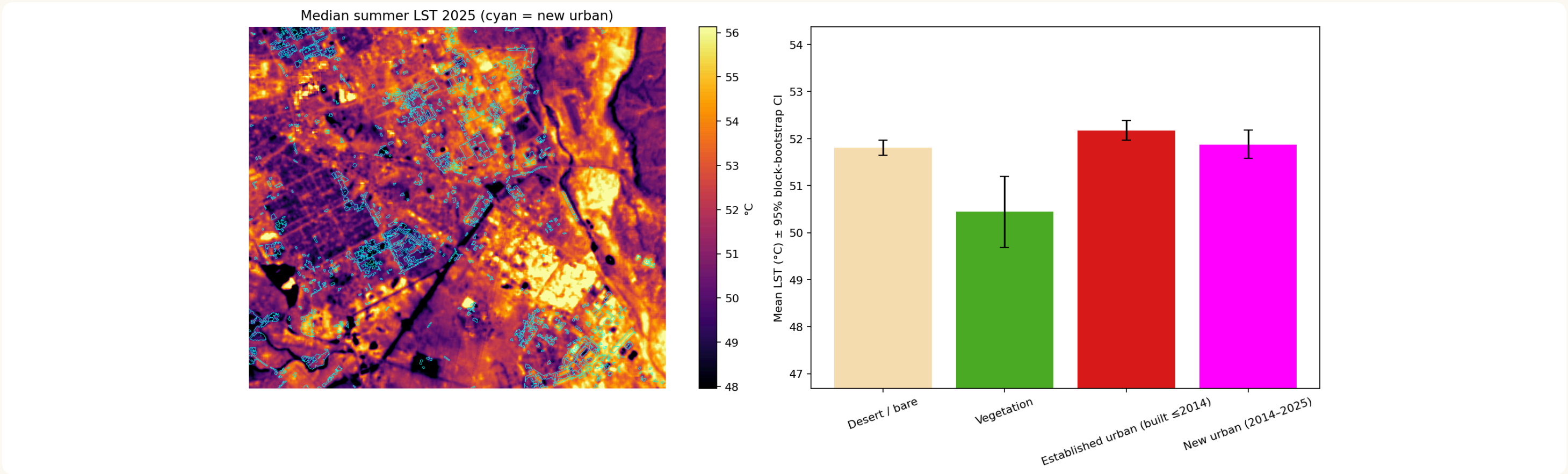
0.86

F1 on an *independent* map (IO LULC 2023)
vs 0.58 NDBI

33 km²

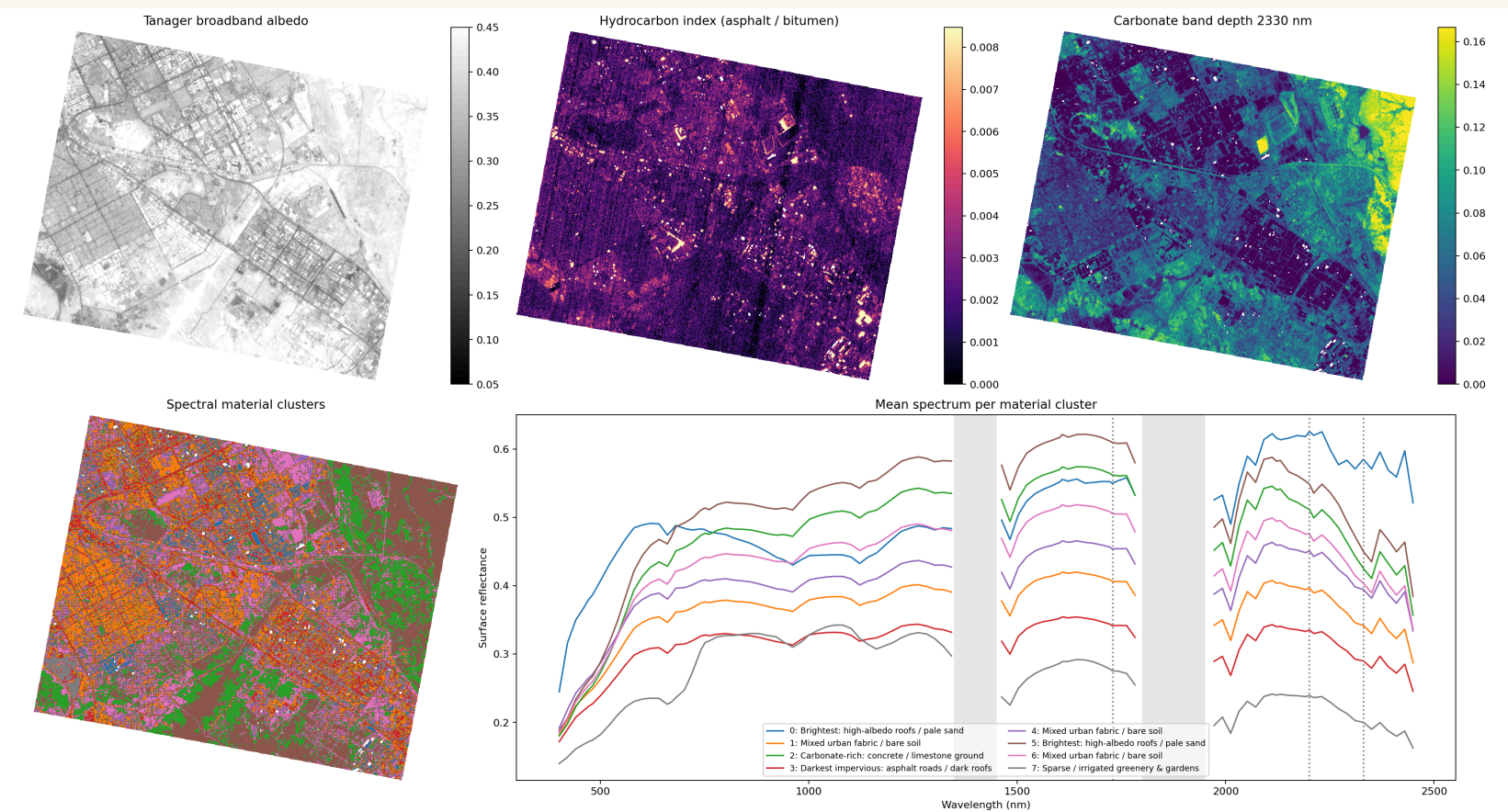
confident new urban land
(p>0.6 after, p<0.4 before)

Peak-summer surface temperature, summer 2025



Established districts **52.2 °C** (95% CI 52.0–52.4) · new districts **51.9 °C** · desert 51.8 °C · vegetation **50.4 °C**. In desert cities, open sand is as hot as the city by day, so the decision-relevant comparison is **between the places where people live**, and greenery is the strongest local coolant.

What makes it hot? Narrow bands see the material



+0.13

R² gain for predicting LST in built-up areas
when Tanager is added (0.31 → 0.44, spatial CV)

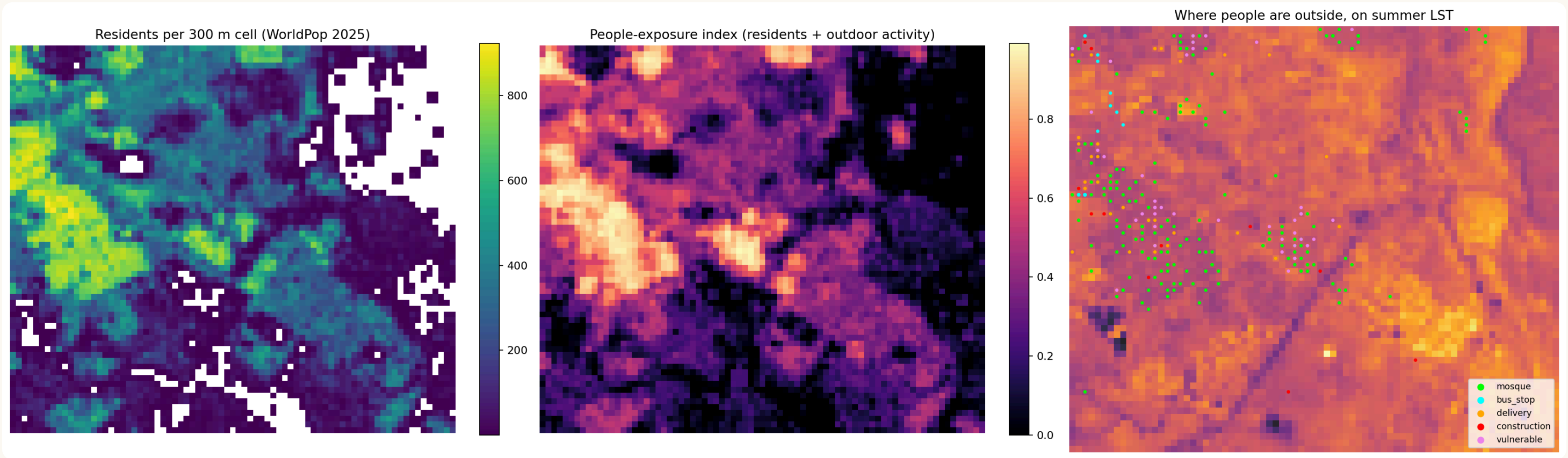
r = 0.87

Tanager vs Landsat albedo
after automatic co-registration

74% → 0.0%

Planet beta mask called bright sand "cloud".
We caught it with a physics test

Who is outside in this heat?

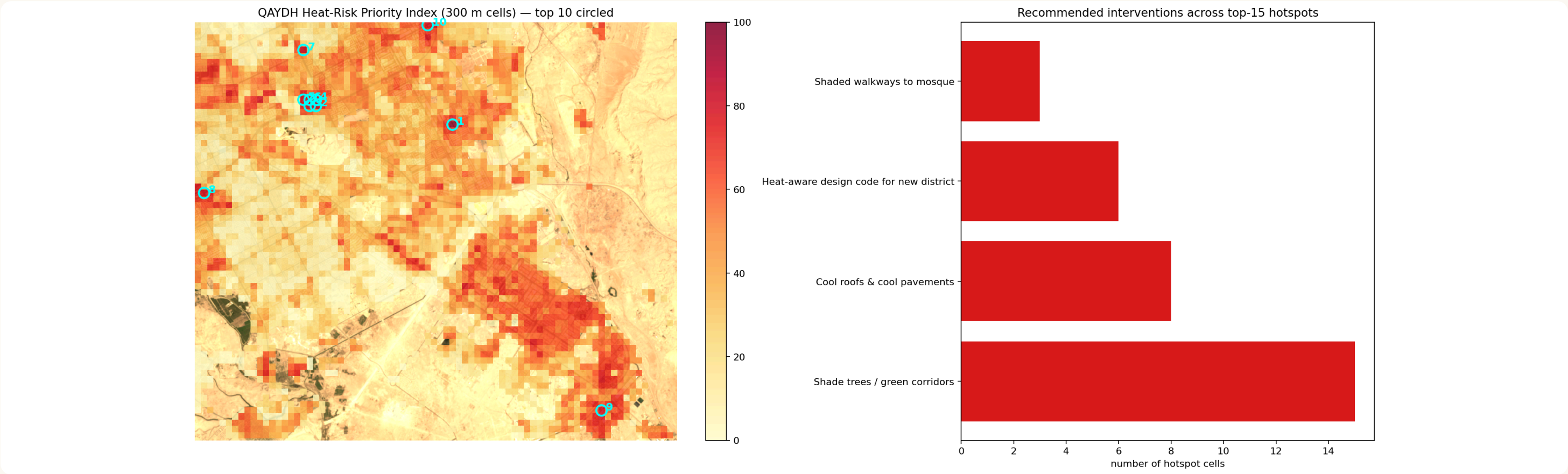


mapped (OSM) per 10,000 residents	hottest 20%	rest of city	
bus stops	0.00	0.63	none mapped
mosques	0.51	1.74	3.4x fewer
parks / gardens	0.13	1.18	9.1x fewer
schools & clinics	0.06	0.45	7.5x fewer

157,076 residents live in the hottest 20% of populated cells.

Counting people instead of buildings changes 90% of the top-20 priorities.

Where to act first, and with what



rank	location	LST	residents	recommended action
#1	24.6220, 46.8640	54.4 °C	403	Shade trees / green corridors + Heat-aware design code for new district
#2	24.6310, 46.7990	55.7 °C	441	Shaded walkways to mosque + Cool roofs & cool pavements + Shade trees / green corridors
#3	24.6310, 46.7960	56.0 °C	307	Cool roofs & cool pavements + Shade trees / green corridors
#4	24.6340, 46.7990	54.2 °C	410	Cool roofs & cool pavements + Shade trees / green corridors
#5	24.6340, 46.7930	55.9 °C	282	Shaded walkways to mosque + Cool roofs & cool pavements + Shade trees / green corridors

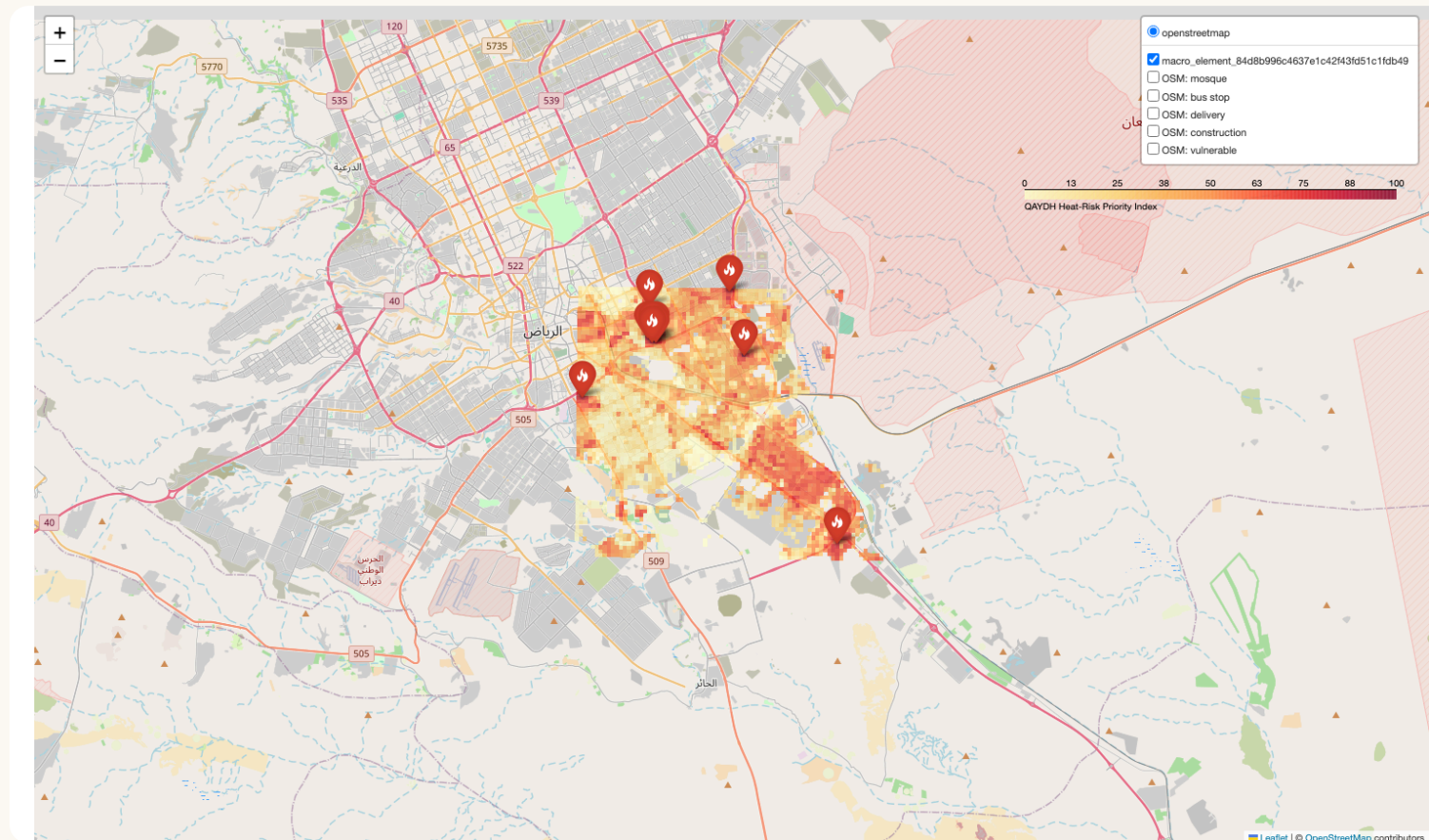
Cool-roof what-if inside built-up areas: **+0.10 albedo** ⇒ **-0.70 °C** (95% CI -0.85 to -0.56), and **+0.10 NDVI** ⇒ **-1.70 °C**, so budgets can be compared in °C.

Every claim has a number and a method

Claim	Metric	Validation	Result
Built-up map beats the official rule	F1 / IoU	5-fold spatial-block CV (1 km) vs WorldCover	0.89 vs 0.58 · IoU 0.81
Generalises to another year & reference	F1	IO LULC 2023, all pixels	0.86 vs 0.58
Detected change is real	precision / F1	new built 2017→23 vs IO LULC	precision 0.64 · F1 0.29
Heat differs between zones	mean LST ± 95% CI	block bootstrap, 300×	52.2 vs veg 50.4 °C
Tanager co-registered	Pearson r	albedo vs Landsat, ±4 px search	0.87
Hyperspectral adds value	ΔR^2 , RMSE	spatial-block CV, ± Tanager	all +0.09 · built-up +0.13
Cool roofs cool	°C per +0.10 albedo	OLS, block-bootstrap CI	-0.70 (-0.85...-0.56)
Priorities are robust	top-20 overlap	4 alternative weightings	60–100%

Honest edge: change detection is precise but conservative (recall 0.19). It only flags confident change, which is what a planner should act on.

How planners use it



- **Heat-risk dashboard:** 300 m cells by priority, top-10 pinned, toggle layers for mosques, bus stops, rider hubs and construction. Hovering shows LST, residents and the action. To be hosted on **glQ**.
- **GeoJSON / REST API:** `GET /cells?city=riyadh&min_hrpi=80` , straight into ArcGIS/QGIS and permitting.
- **Annual heat brief + alerts:** after each summer, a one-page brief per municipality flags new districts entering the top decile, and projects that moved a cell out of it.

Value for cities, people and budgets

Impact

- Protects people outdoors: worshippers, bus riders, riders, workers, children
- Targets the shade and cooling budget where it lowers the most °C per dirham
- SDG 3 · 11 · 13 · UAE Net Zero 2050 · Saudi Green Riyadh

Who pays

- Municipalities & planning departments (annual city licence)
- Real-estate developers (heat-aware master-plan check)
- Delivery platforms & contractors (rider and worker heat-safety planning)

Why it scales

- Open data only: any Arab city with Landsat + one hyperspectral scene
- Runs in ~20 min per city per year
- Satellite 813 adds regional hyperspectral revisit

From PoC to MVP in incubation

Incubation plan

1. UAE cities (Abu Dhabi, Dubai, Al Ain) with **Satellite 813 / MBZ-SAT** data
2. Worker-housing and delivery-hub layers from municipal and platform partners
3. Calibrate LST with NCM / municipal weather stations. Field-check 200–300 material points
4. Dashboard on **gIQ** with a what-if slider (roof albedo, trees, shaded stops)

Known limits

- LST is a ~10:30 surface snapshot, not afternoon air temperature
- Reference maps, not field truth. Sandy low-density suburbs are hardest
- OSM under-maps labour housing and rider waiting spots
- Cooling what-if is a statistical slope with a CI, not a simulation
- One Tanager date (May). Materials are stable, shade is not

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Know where it burns, why, and who is outside.

157,076 residents in the hottest 20% of east Riyadh · +0.13 R^2 from hyperspectral · -0.70 °C per +0.10 roof albedo

Code, notebook and outputs: [GitHub repository \(README\)](#) · Data © Planet Labs PBC (CC-BY-4.0), USGS, ESA, Impact Observatory, WorldPop, OpenStreetMap contributors